HopTF Mutant Transport Analysis
Generated: May 11, 2026

Question: do curated TF sequence mutations change the HopTF-predicted transport endpoint in the expected direction?

Inputs: ESM-C 600M mean_non_special embeddings were generated on laplace for six verified mutant isoform sequences. The strict frozen endpoint evaluator completed successfully: 6 / 6 verified mutants predicted.

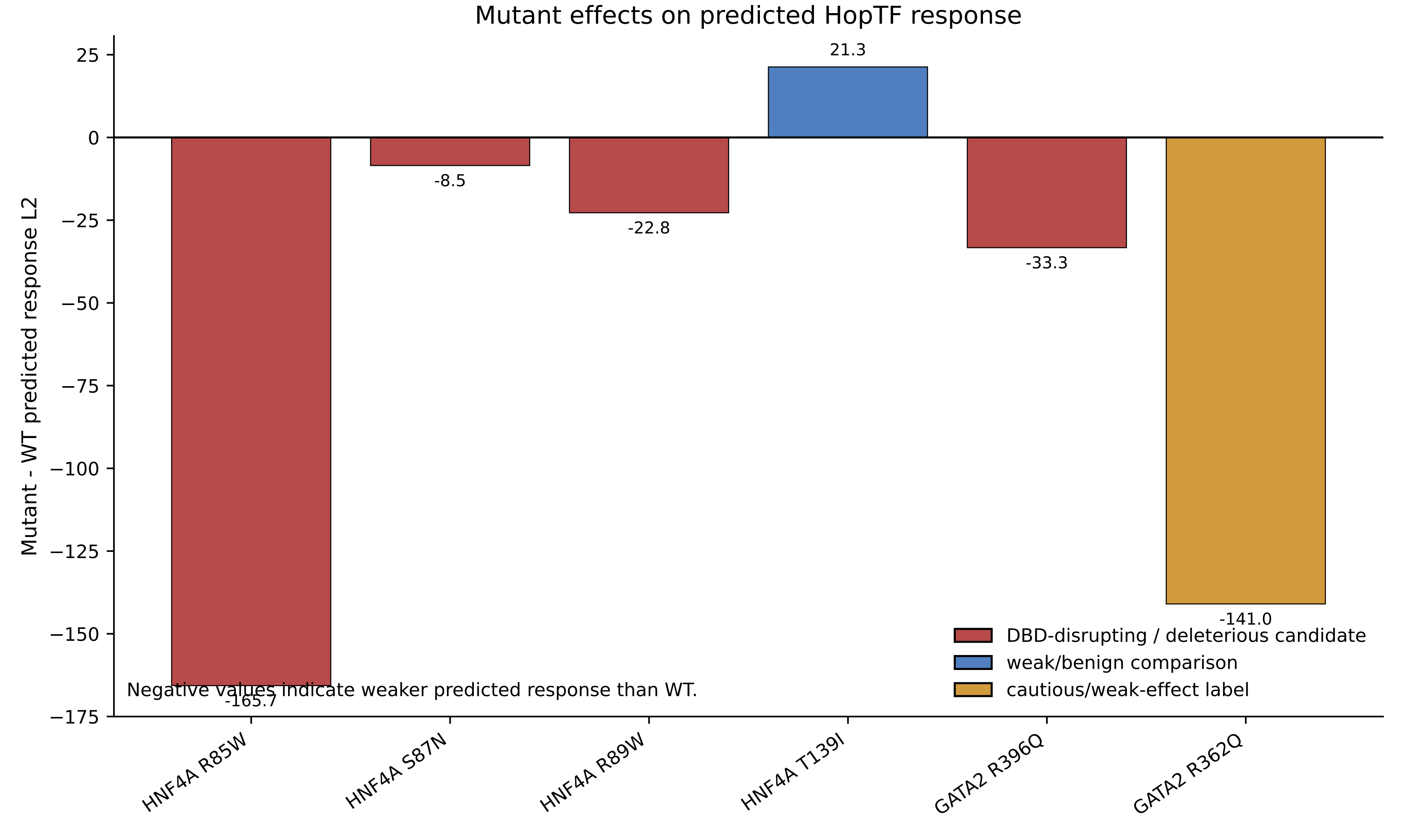
Main readout: mutant_minus_wt_predicted_response_l2 = mutant predicted response L2 - WT predicted response L2. Negative values mean weaker predicted response than WT.

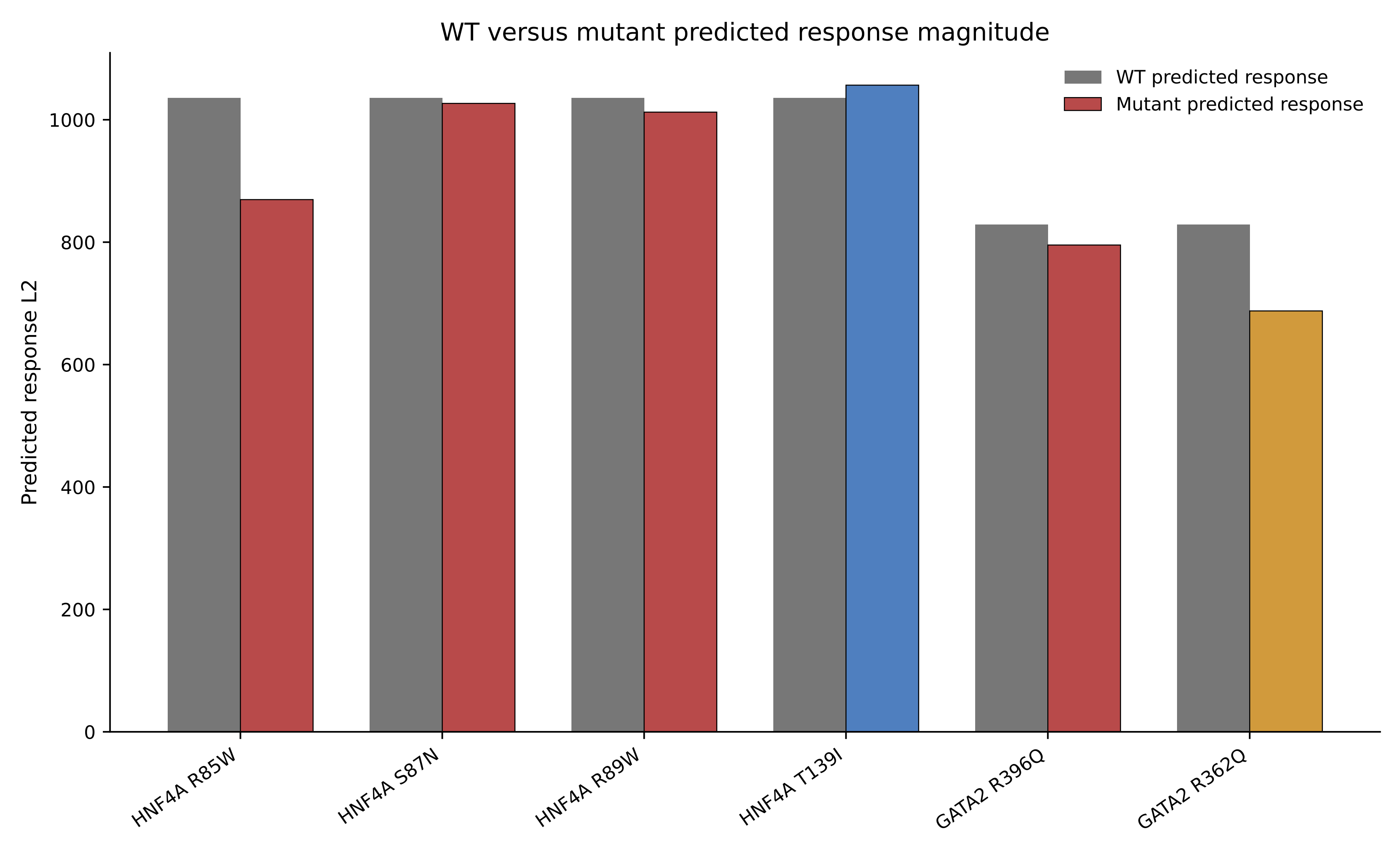
Headline result: HNF4A is directionally supportive. The three DNA-binding-disrupting substitutions all reduce predicted response relative to WT: R85W (-165.66), S87N (-8.49), and R89W (-22.76). The HNF4A weak/benign comparison T139I slightly increases predicted response (+21.30), which is consistent with using it as a contrast rather than a loss-of-function mutation.

GATA2 is directionally supportive but should be interpreted cautiously. R396Q reduces predicted response modestly (-33.31). R362Q reduces predicted response strongly (-140.98), but it was already labeled as a cautious/weak-effect comparison, so this should not be overclaimed.

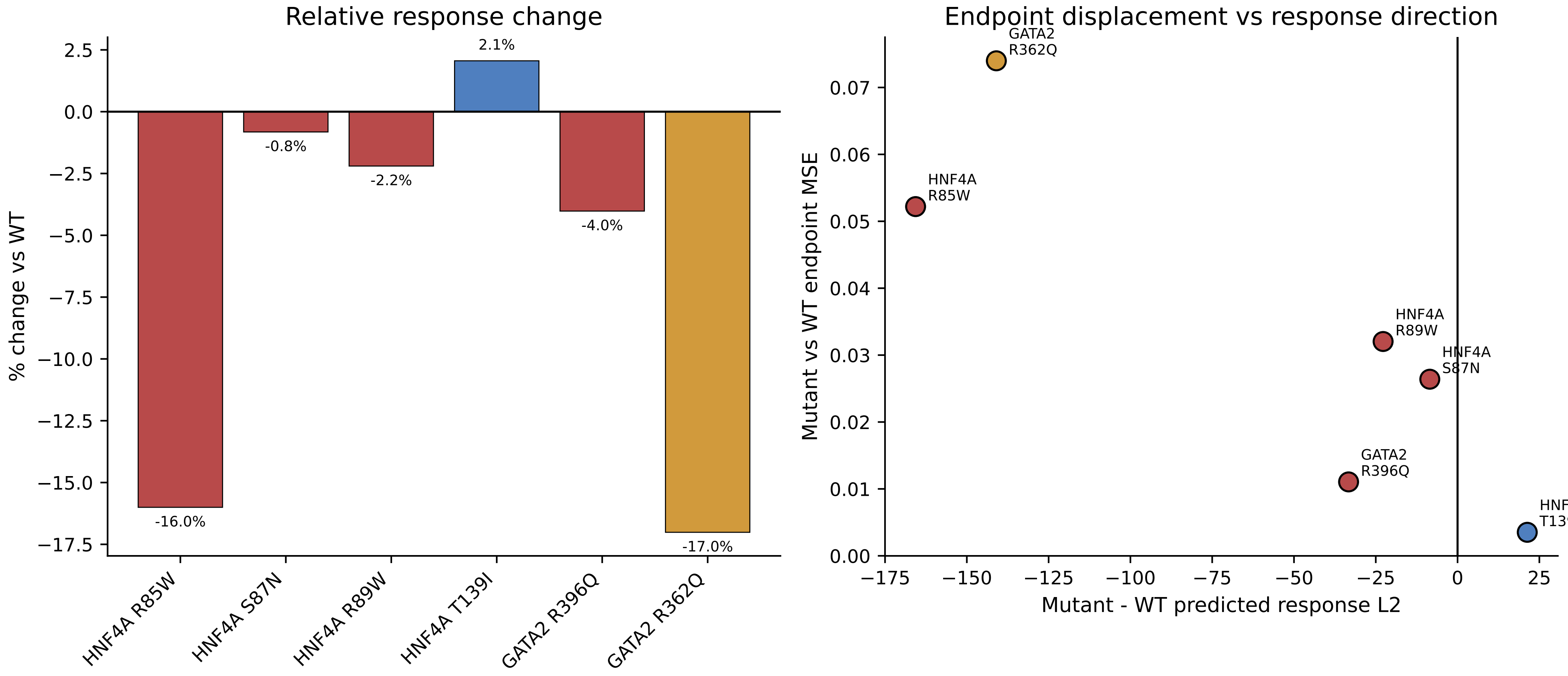
Interpretation: this is a preliminary sequence-sensitivity result for a frozen HopTF endpoint model, not full biological validation. The endpoint is a perturbation-level PCA centroid transport target, and the natural same-gene isoform panel remains imperfect (8 / 14 stable label-aware passes). The result is strongest as evidence that the sequence-conditioned model responds nontrivially to meaningful amino-acid substitutions.

Traceability: predictions from tmp/laplace_returns/hoptf_mutant_transport_laplace_20260511_140909/mutant_endpoint_predictions.csv; report metadata from tmp/laplace_returns/hoptf_mutant_transport_laplace_20260511_140909/mutant_endpoint_predictions_report.json.


Mutant effects on predicted HopTF response
Mutant - WT predicted response L2
25
0
−25
−50
−75
−100
−125
−150
−175
21.3
-8.5
-22.8
-33.3
-141.0
-165.7
DBD-disrupting / deleterious candidate
weak/benign comparison
cautious/weak-effect label
Negative values indicate weaker predicted response than WT.
HNF4A R85W
HNF4A S87N
HNF4A R89W
HNF4A T139I
GATA2 R396Q
GATA2 R362Q



WT versus mutant predicted response magnitude
WT predicted response
Mutant predicted response
Predicted response L2
0
200
400
600
800
1000
HNF4A R85W
HNF4A S87N
HNF4A R89W
HNF4A T139I
GATA2 R396Q
GATA2 R362Q



Relative response change
% change vs WT
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
2.1%
-0.8%
-2.2%
-4.0%
-16.0%
-17.0%
HNF4A R85W
HNF4A S87N
HNF4A R89W
HNF4A T139I
GATA2 R396Q
GATA2 R362Q
Endpoint displacement vs response direction
Mutant vs WT endpoint MSE
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
GATA2
R362Q
HNF4A
R85W
HNF4A
R89W
HNF4A
S87N
GATA2
R396Q
HNF4A
T139I
−175
−150
−125
−100
−75
−50
−25
0
25
Mutant - WT predicted response L2

Numerical summary and caveats

Gene	Mutation	Class	WT L2	Mutant L2	Delta L2	% change	Endpoint MSE
HNF4A	R85W	DBD-disrupting	1035.217	869.555	-165.662	-16.003	0.052
HNF4A	S87N	DBD-disrupting	1035.217	1026.729	-8.487	-0.820	0.026
HNF4A	R89W	DBD-disrupting	1035.217	1012.458	-22.759	-2.198	0.032
HNF4A	T139I	weak/benign compariso	1035.217	1056.519	21.303	2.058	0.004
GATA2	R396Q	deleterious candidate	828.636	795.323	-33.312	-4.020	0.011
GATA2	R362Q	cautious/weak-effect lab	828.636	687.658	-140.977	-17.013	0.074

Caveats: endpoint predictions use a frozen smoke-model checkpoint over perturbation-level PCA centroid endpoints. The natural same-gene isoform benchmark is only partially stable, so mutant transport should be framed as sequence sensitivity rather than definitive biological validation.
TP53 mutations remain excluded because local isoform coordinates do not match canonical residues.